G0251 FOUNDATION OF BUSINESS ANALYTICS
TUTORIAL 4

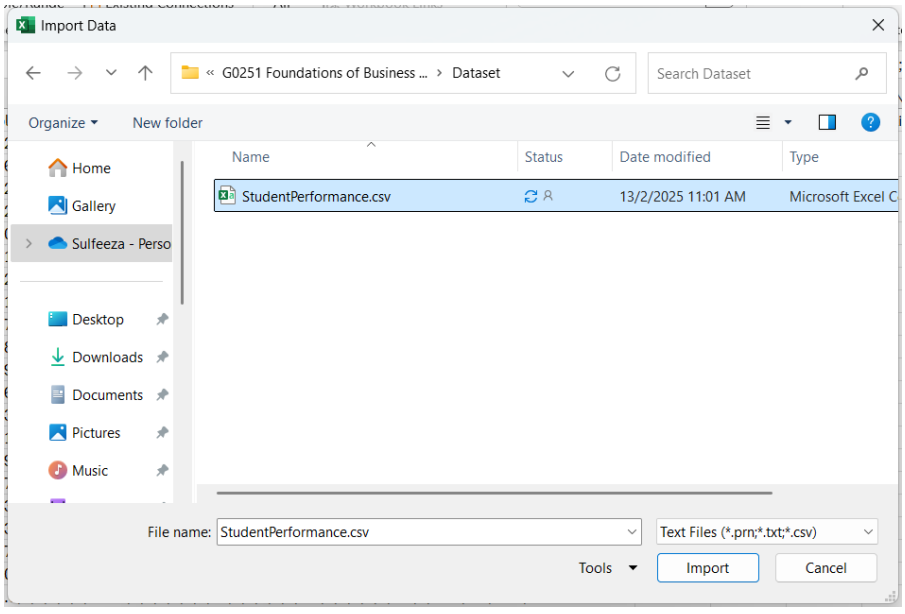
Prepared by:
Sulfeeza binti Mohd Drus
School of Economics and Management (SEM)

Analyzing University Students' Performance

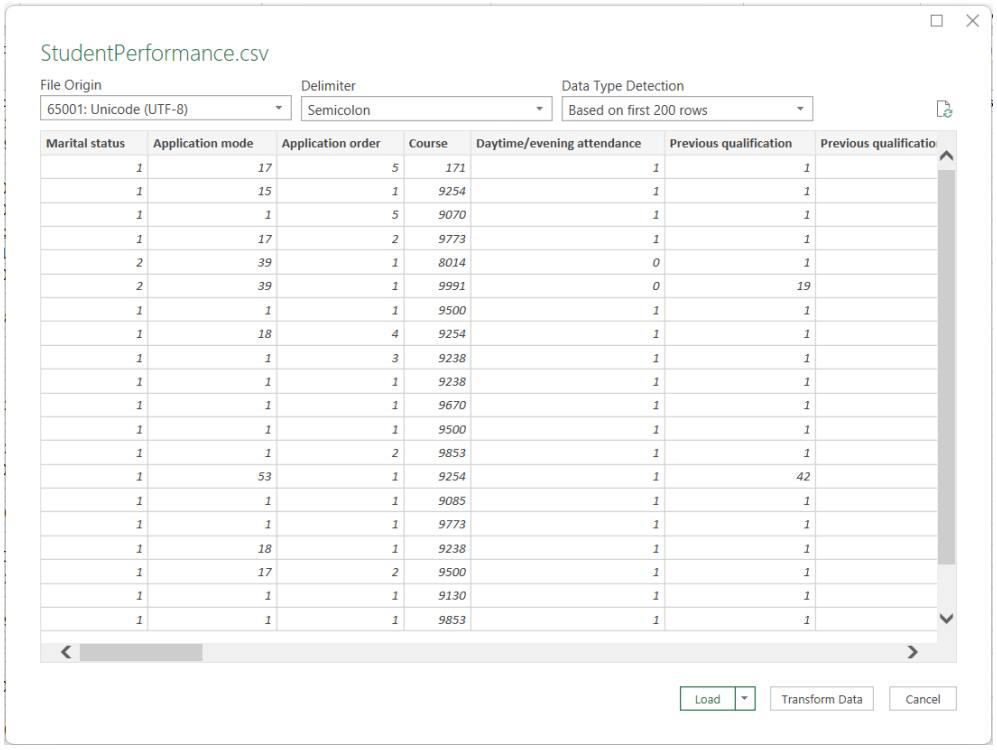
The dataset contains information about students' performance and their status of study (either as dropout, enrolled or graduate).

Converting Comma-Separated (.csv) file to Excel (.xlsx) file

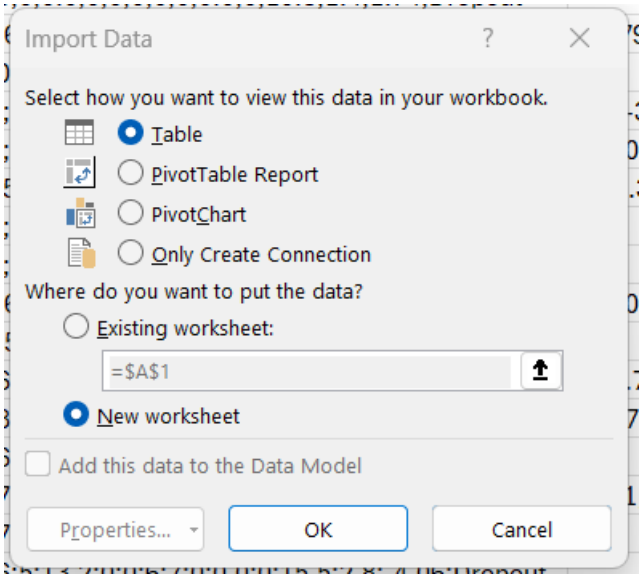
1. Download the file StudentPerformance.csv into your own laptop.
2. Open the StudentPerformance.csv file using Microsoft Excel.
3. Go to the Data tab.
4. In the Get & Transform Data group, click at the Get Data from CSV command.
5. A window will pop-up as below:



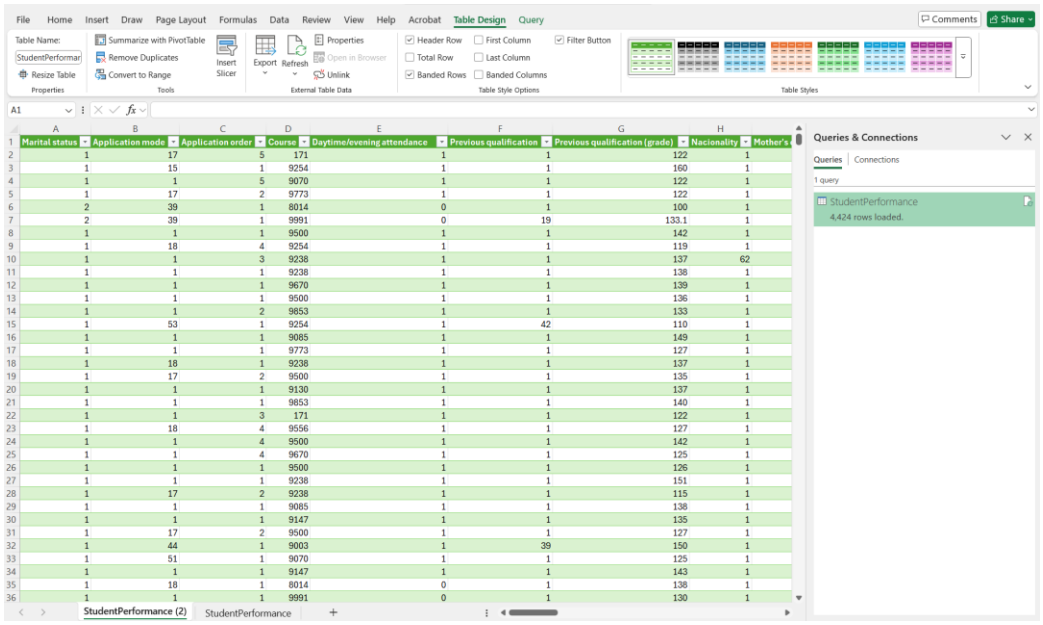
6. Choose the location in your computer where you save your StudentPerformance.csv earlier and press the button Import.
7. A Power Query Editor will be opened as below:



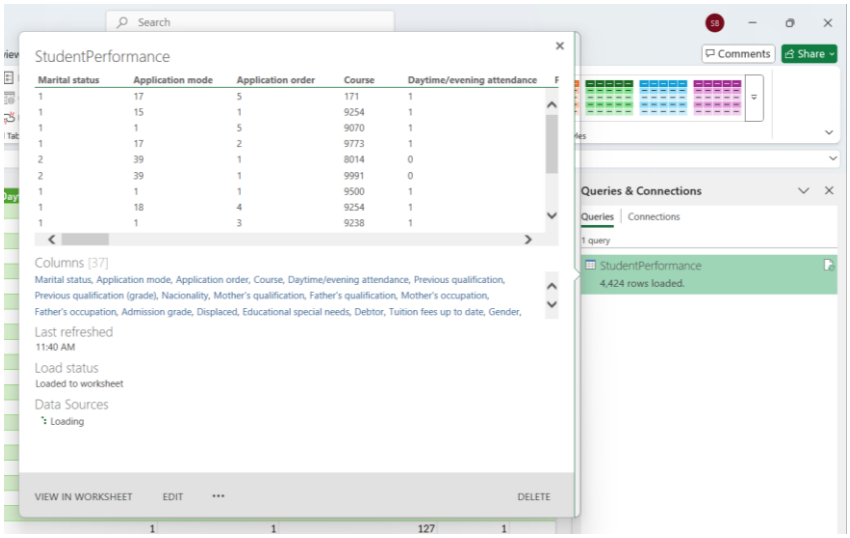
8. The Power Query Editor will display Data Preview - a grid view of the data, similar to a table format, displaying all rows and columns of the imported data.
9. Then click the Drop-Down arrow at Load button and click the Load To option.
10. The Import Data dialog box will be opened as below:



11. Choose the default options and click OK.
12. This will load a new table on a new worksheet in the same file (as below):



13. Mouse over to StudentPerformance at Queries and Connections and the following window will be displayed. It shows that this dataset contains 4424 rows and 37 columns.



14. Go to the File tab and save this file as an Excel file using the same name – StudentPeformance.

Understanding the dataset

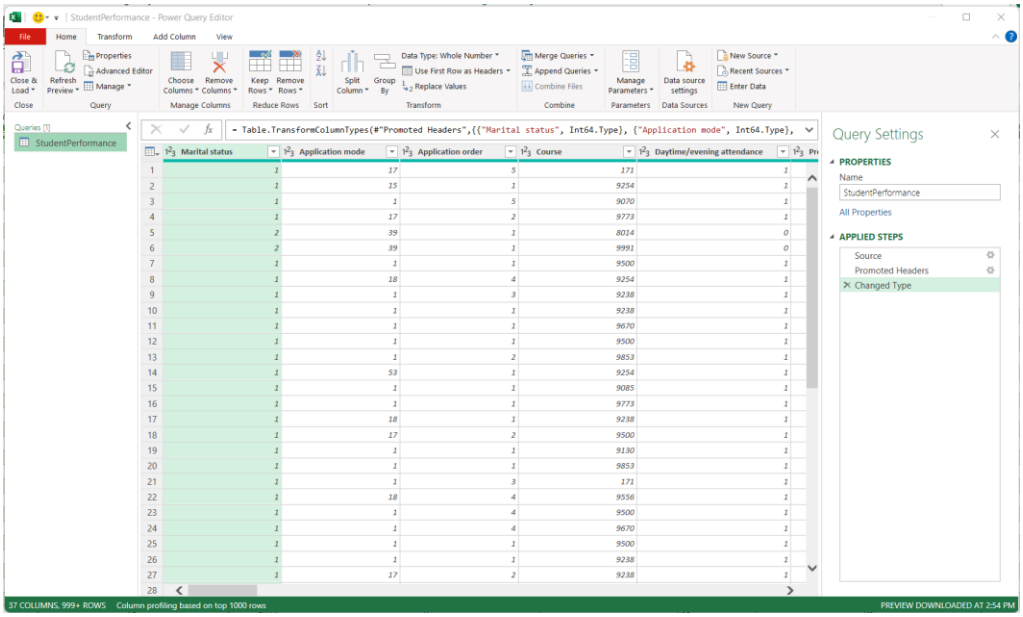
Here is the list of columns in the dataset:

Column Name	Description	Value
Marital Status	Marital Status	1 – Single 2 – Married 3 – Widower 4 – Divorced 5 – Facto Union 6 – Legally Separated
Application mode	How the application is being made	To be removed
Application order	The ranking of the application	0 – 1st choice 1 – 2nd choice till 9 – Last choice
Course	Code for the course registered	33 - Biofuel Production Technologies 171 - Animation and Multimedia Design 8014 - Social Service (evening attendance) 9003 - Agronomy 9070 - Communication Design 9085 - Veterinary Nursing 9119 - Informatics Engineering 9130 - Equinculture 9147 - Management 9238 - Social Service 9254 - Tourism 9500 - Nursing 9556 - Oral Hygiene 9670 - Advertising and Marketing Management 9773 - Journalism and Communication 9853 - Basic Education 9991 - Management (evening attendance)
Daytime/evening attendance	Class Attendance	0 - Evening 1 – Daytime
Previous qualification	Education Level	To be removed
Previous qualification (grade)	Grade of previous qualification	Between 0 and 200
Nacionality	Nationality	1 – Portuguese 2 – German 6 – Spanish 11 – Italian 13 – Dutch 14 – English 17 – Lithuanian 21 – Angolan 22 - Cape Verdean 24 – Guinean 25 – Mozambican 26 – Santomean 32 – Turkish 41 – Brazilian 62 – Romanian 100 – Republic of Moldova

Column Name	Description	Value
		101 – Mexican 103 – Ukrainian 105 – Russian 108 – Cuban 109 – Colombian
Mother's qualification	Education Level	To be removed
Father's qualification	Education Level	To be removed
Mother's occupation	Occupation	To be removed
Father's occupation	Occupation	To be removed
Admission grade	Admission grade	Between 0 and 200
Displaced		To be removed
Educational special needs		To be removed
Debtor		To be removed
Tuition fees up to date		To be removed
Gender	Gender	0 – Female 1 – Male
Scholarship holder		0 – No 1 – Yes
Age at enrollment	Age of student at enrollment	
International	Is the student an international student?	0 – No 1 – Yes
Curricular units 1st sem (credited)		To be removed
Curricular units 1st sem (enrolled)	Number of curricular units enrolled in the 1st semester	
Curricular units 1st sem (evaluations)		To be removed
Curricular units 1st sem (approved)		To be removed
Curricular units 1st sem (grade)	Grade average in the 1st semester	Between 0 and 20
Curricular units 1st sem (without evaluations)		To be removed
Curricular units 2nd sem (credited)		To be removed
Curricular units 2nd sem (enrolled)	Number of curricular units enrolled in the 2nd semester	
Curricular units 2nd sem (evaluations)		To be removed
Curricular units 2nd sem (approved)		To be removed
Curricular units 2nd sem (grade)	Grade average in the 2nd semester	Between 0 and 20
Curricular units 2nd sem (without evaluations)		To be removed
Unemployment rate	The % of unemployment rate	
Inflation rate	The rate of inflation	
GDP	Gross Domestic Product	GDP
Target	The status of the student at the end of the course	Dropout Enrolled Graduate

Removing unwanted columns

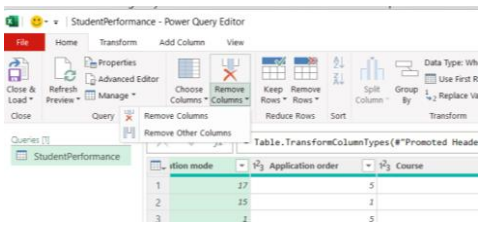
1. Go to Query tab and then click at the Edit command under the Edit Group.
2. A Power Query Editor will be displayed as below:



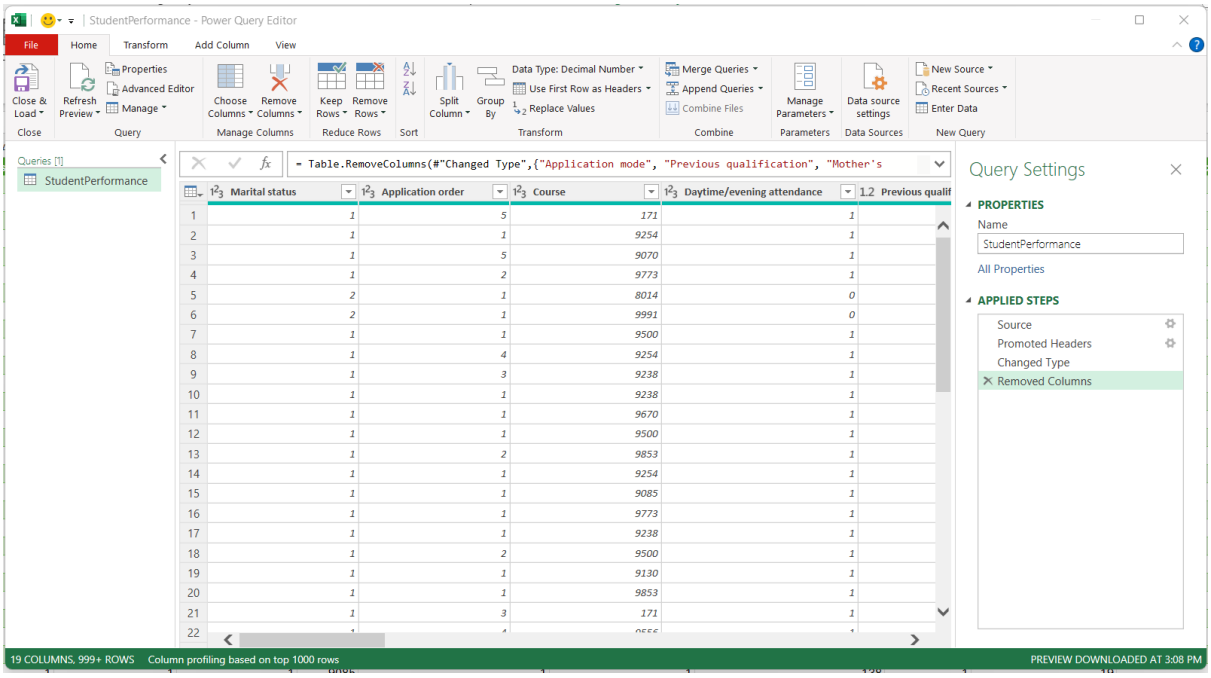
3. While pressing the Ctrl key, select the columns listed below.

- Application mode
- Previous qualification
- Mother's qualification
- Father's qualification
- Mother's occupation
- Father's occupation
- Displaced
- Educational special needs
- Debtor
- Tuition fees up to date
- Curricular units 1st sem (credited)
- Curricular units 1st sem (evaluations)
- Curricular units 1st sem (approved)
- Curricular units 1st sem (without evaluations)
- Curricular units 2nd sem (credited)
- Curricular units 2nd sem (evaluations)
- Curricular units 2nd sem (approved)
- Curricular units 2nd sem (without evaluations)

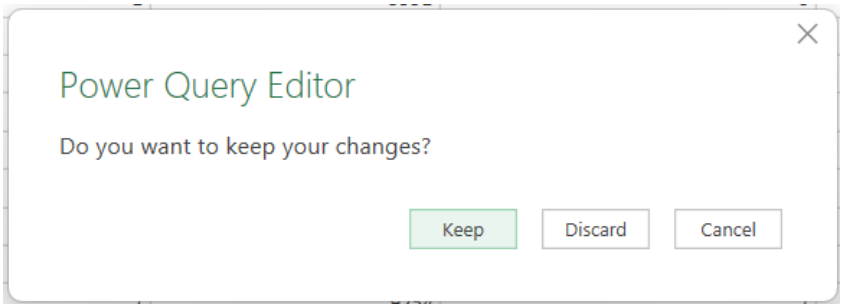
4. Then click at the Remove Columns.



5. Once you have removed all these 18 columns, you can see that your dataset only has 19 columns.

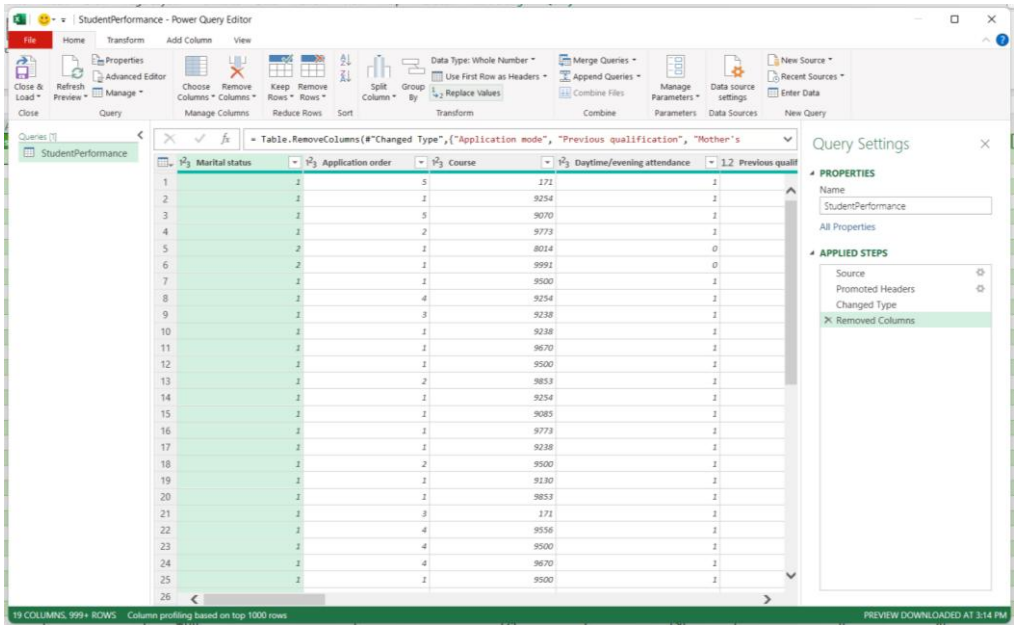


6. Close the Power Query Editor and click on the Keep button (as shown below):

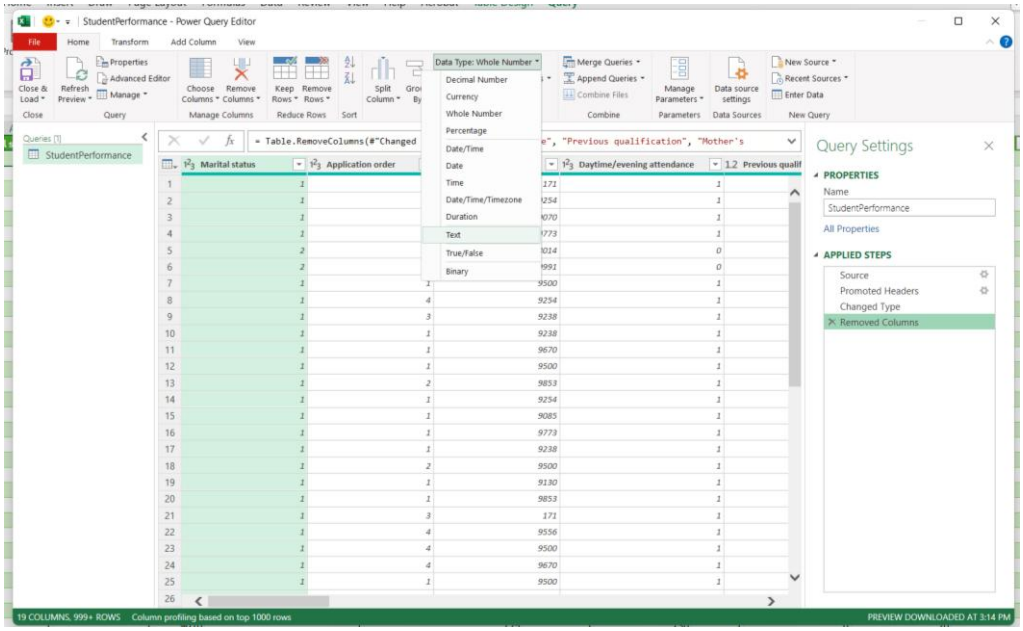


Replacing Values for Marital Status column

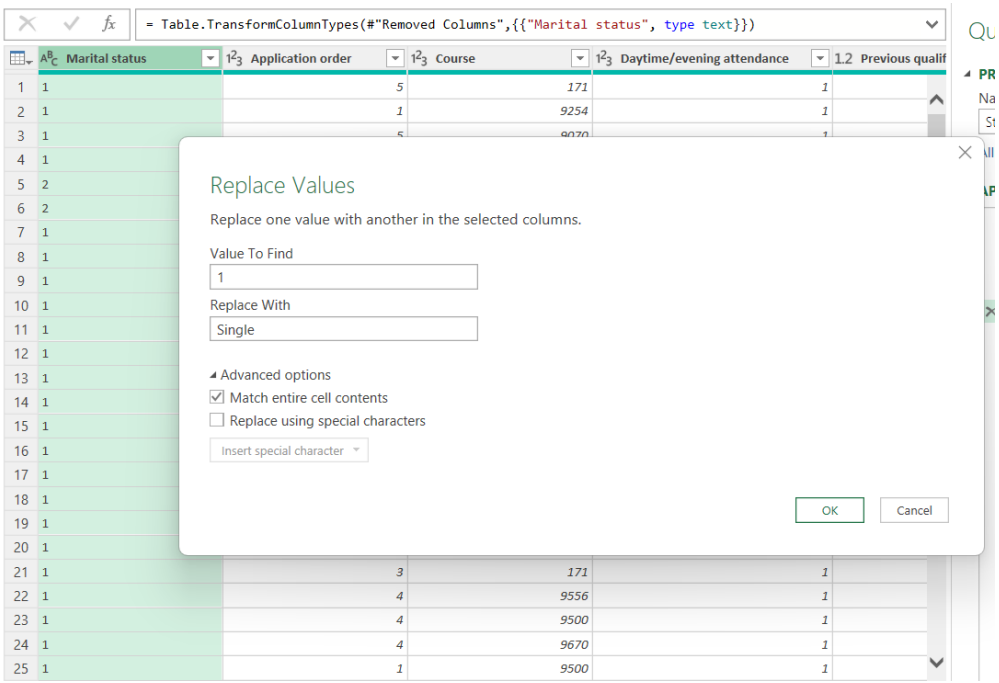
1. We want to replace the numerical values representing the marital status in column A to the textual values, such as 1 will be replaced with Single, 2 will be replaced with Married, 3 will be replaced with Widower, 4 will be replaced with Divorced, 5 will be replaced with Facto Union and 6 will be replaced with Legally Separated.
2. Go to Query tab and then click at the Edit command under the Edit Group.
3. A Power Query Editor will be displayed as below:



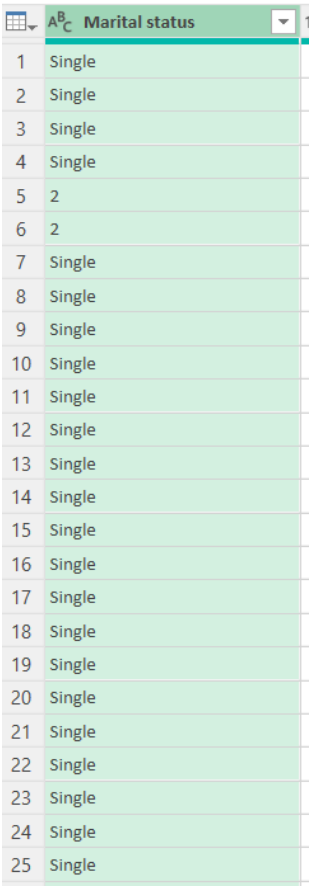
4. While selecting the Marital Status column, click on the Data Type in the Transform group. We need to change the original data type of Marital Status from Whole Number to Text. After you change this, the data type will be changed to Text (as shown below).



5. Once the data type has been changed to Text, then click on the Replace Value in the Transform group. A dialog box will be displayed as follows and enter the following details.



6. Once you have successfully replaced for Single, the values in Marital Status column will be as follows:



7. Now change for the other values as stated in Step 1.

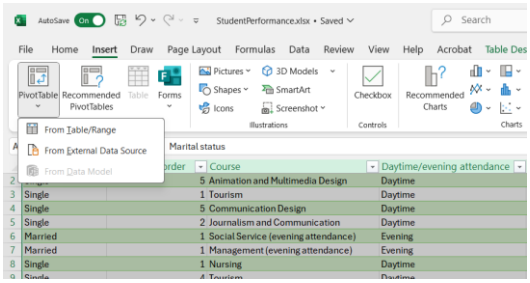
Replacing Values for other columns

For the following columns, changes the values in the columns based on the respective values shown below. Use the same steps that we have done earlier.

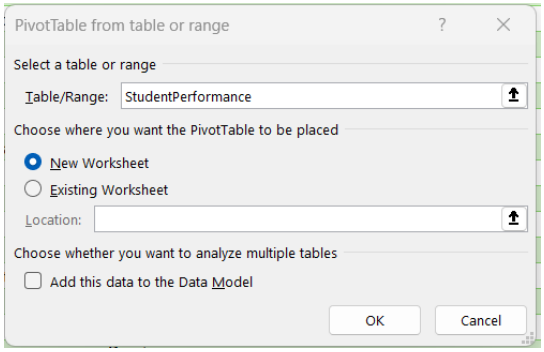
Column Name	Value
Course	33 - Biofuel Production Technologies 171 - Animation and Multimedia Design 8014 - Social Service (evening attendance) 9003 - Agronomy 9070 - Communication Design 9085 - Veterinary Nursing 9119 - Informatics Engineering 9130 - Equinculture 9147 - Management 9238 - Social Service 9254 - Tourism 9500 - Nursing 9556 - Oral Hygiene 9670 - Advertising and Marketing Management 9773 - Journalism and Communication 9853 - Basic Education 9991 - Management (evening attendance)
Daytime/evening attendance	0 - Evening 1 – Daytime
Nacionality	1 – Portuguese 2 – German 6 – Spanish 11 – Italian 13 – Dutch 14 – English 17 – Lithuanian 21 – Angolan 22 – Cape Verdean 24 – Guinean 25 – Mozambican 26 – Santomean 32 – Turkish 41 – Brazilian 62 – Romanian 100 – Moldovan 101 – Mexican 103 – Ukrainian 105 – Russian 108 – Cuban 109 – Colombian
Gender	0 – Female 1 – Male
Scholarship holder	0 – No 1 – Yes
International	0 – No 1 – Yes

Creating Bar Chart using PivotTable

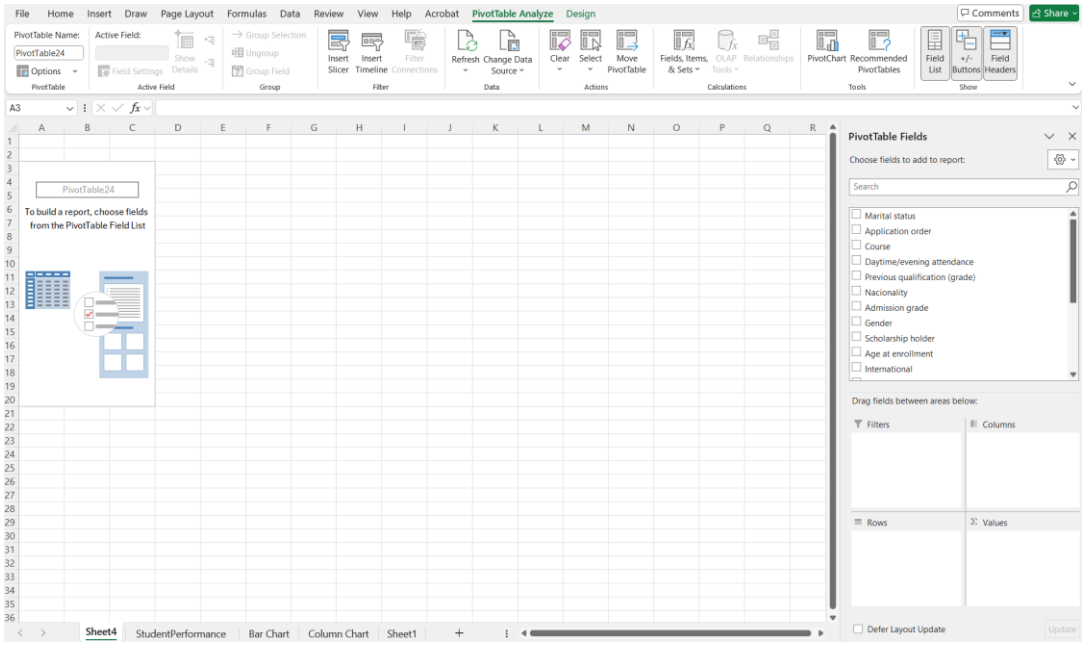
1. Press Ctrl+A to select all the cells that contains data in the StudentPerformance tab.
2. Then, click at the Insert tab and select Pivot Table under the Tables command.



3. A dialog box will be opened (as below). Click OK.

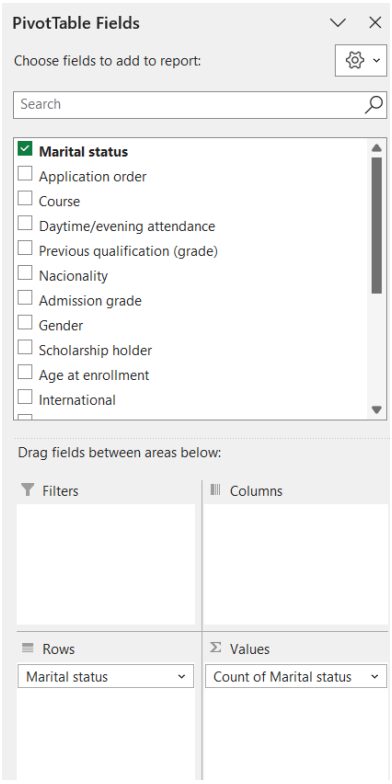


4. A new worksheet will be opened.



5. The PivotTable Fields Pane (on the right) contains all the columns that we have in our dataset.

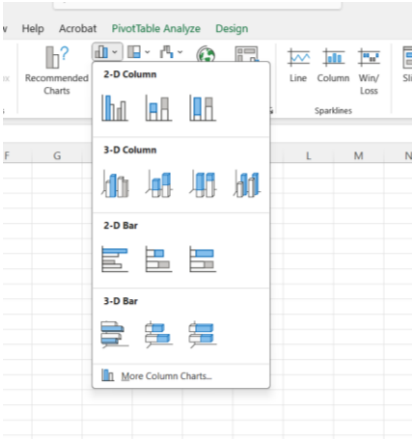
6. Drag Marital Status into the Rows box and Values box.



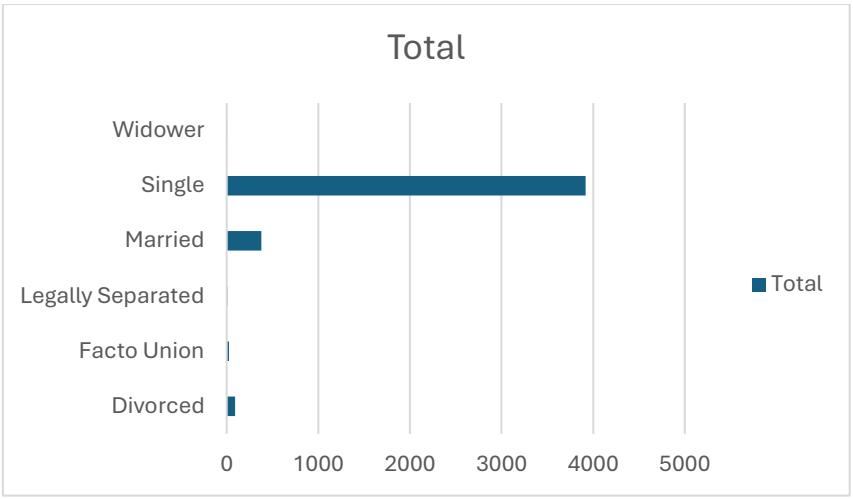
7. A PivotTable will be created.

Row Labels	Count of Marital status
Divorced	91
Facto Union	25
Legally Separated	6
Married	379
Single	3919
Widower	4
Grand Total	**4424**

8. While selecting the content of the PivotTable, go to the Insert Table and choose Bar Chart under the Charts group.



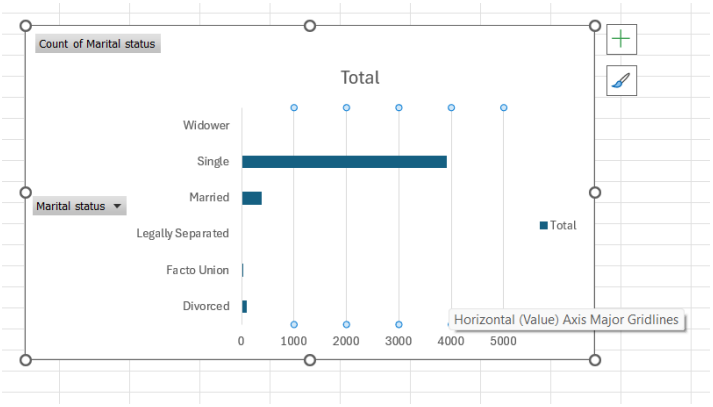
9. Select the 2D Bar chart and the chart (as below) will be created.



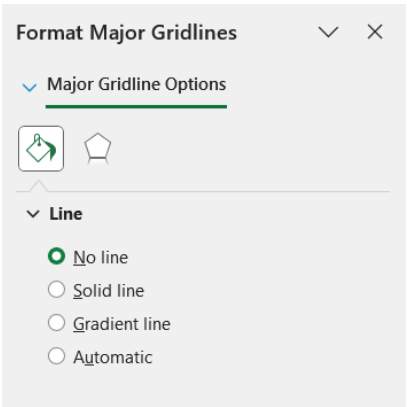
10. Now, we want to improve the look of our bar chart by removing the unnecessary "ink" and the scale of measurement.

    a. <u>Removing the gridlines</u>

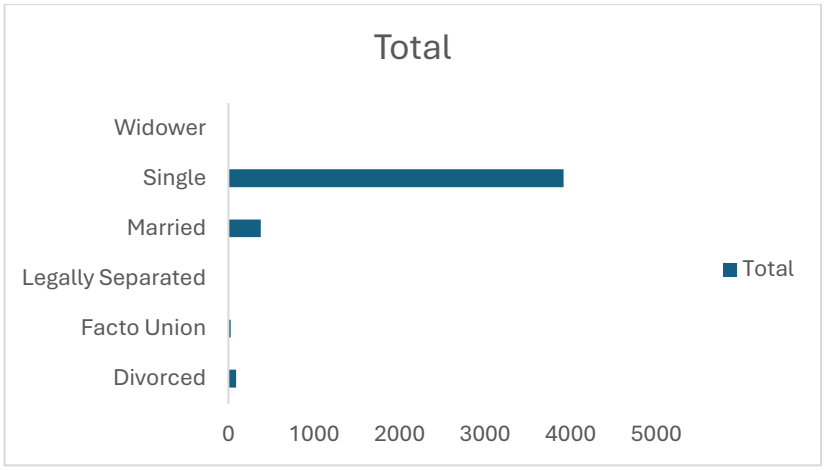
        1. Move your mouse to the grid line so that the lines will be highlighted



        2. Right-click and choose Format Gridlines. The Format Gridlines Pane will be opened. Choose No line.

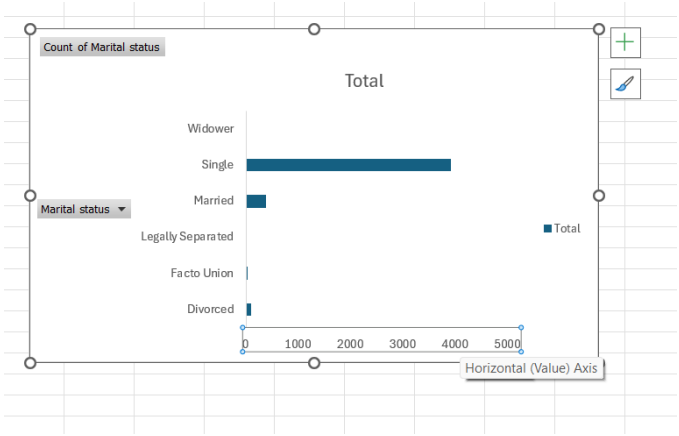


        3. Now the gridlines will not be displayed anymore on the chart.

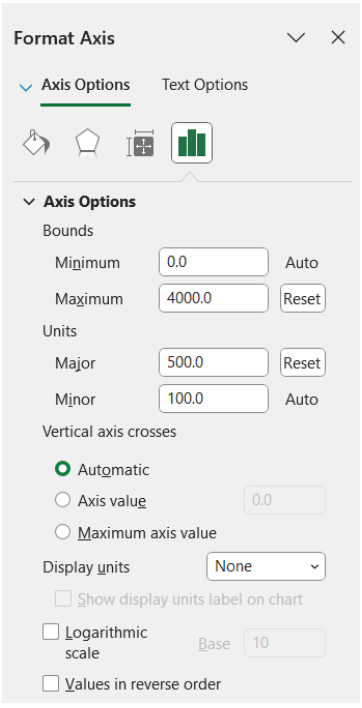


b. Re-scaling the values on the axis

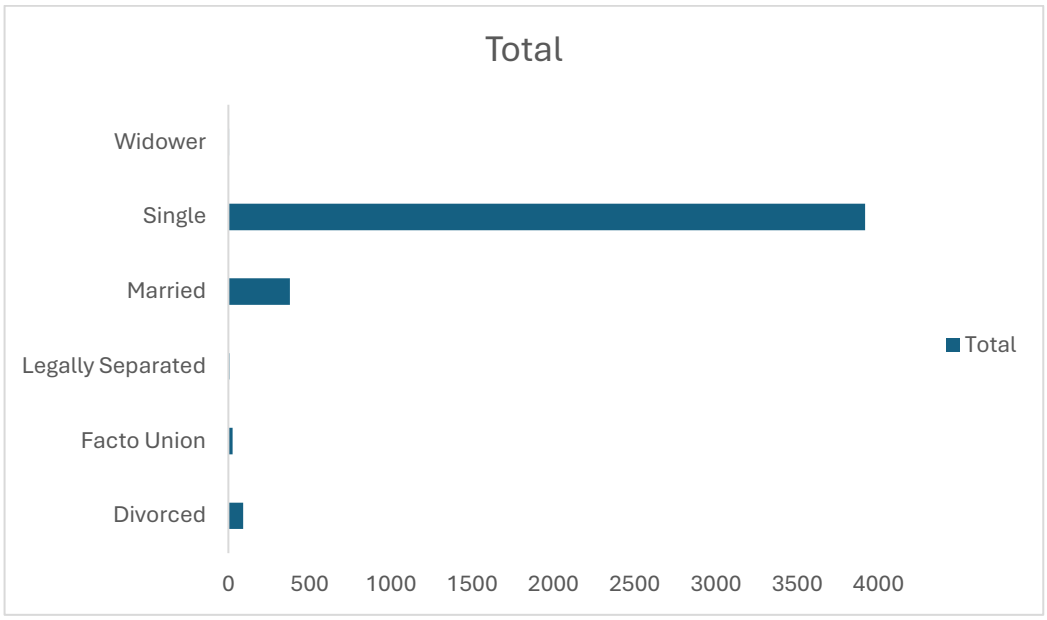
1. Move your mouse to the horizontal value axis so that the values will be highlighted



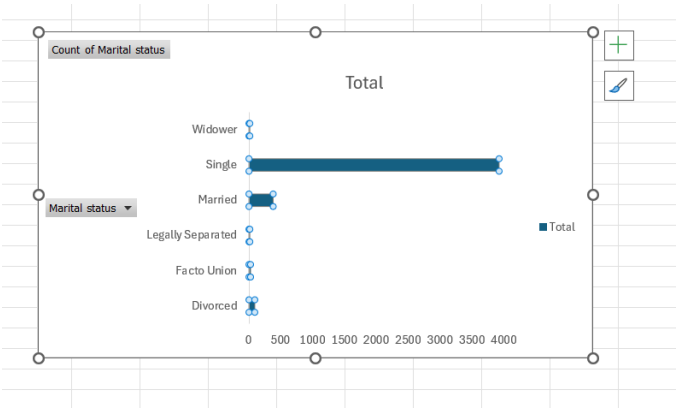
2. Right-click and choose Format Axis. The Format Axis Pane will be opened. Make the following changes.



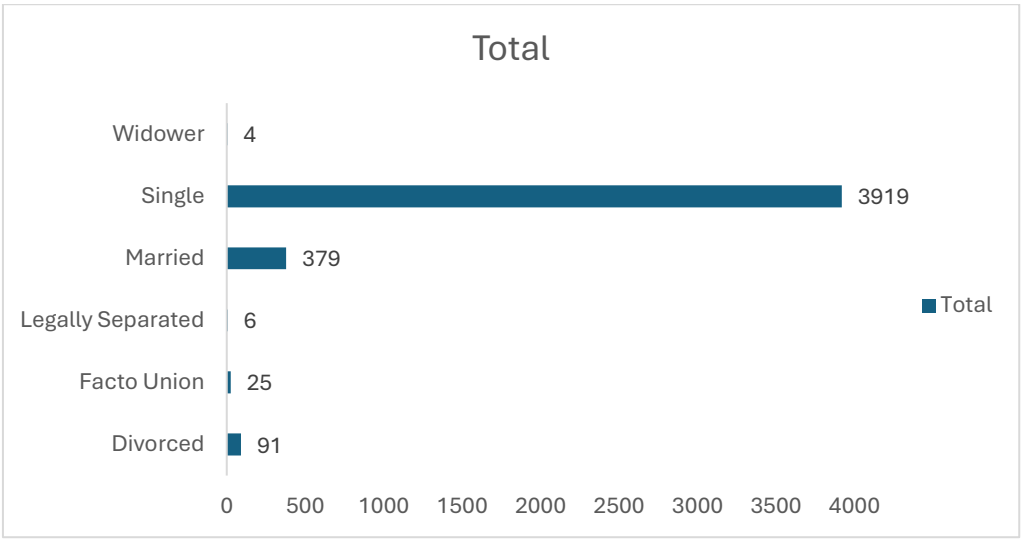
3. Now the values on the axis will be changed accordingly.



11. Let's say now we want to include the count for each marital status category in the chart.
12. Move your mouse to the bar so that the bars will be highlighted

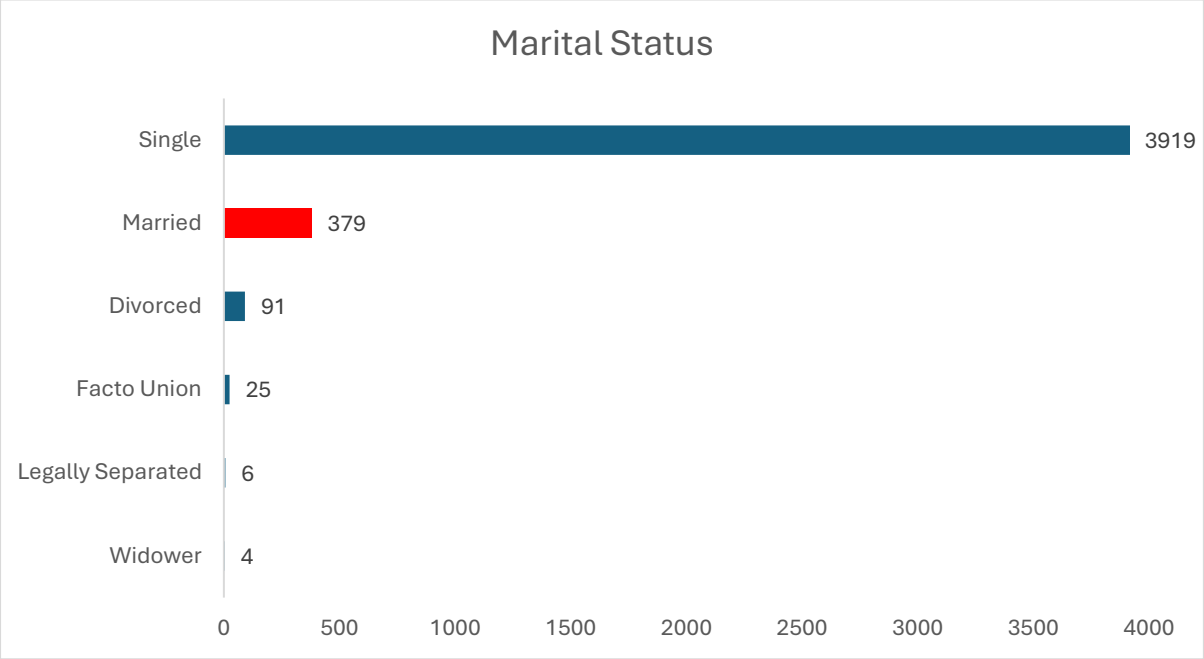


13. Right-click and choose Add Data Labels. The value for each bar will be displayed.

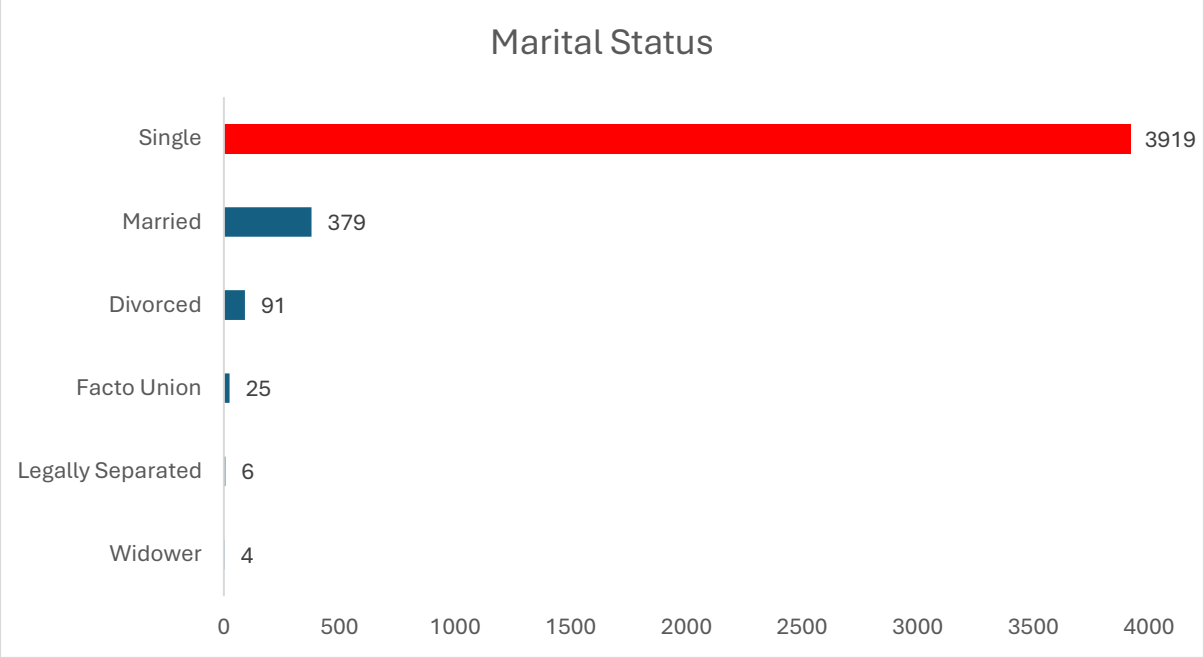


Try On Your Own

1. Use different colour for the highest value in marital status



2. Sort categories for clarity



Creating Stacked Column Chart using PivotTable

1. There are four boxes in the PivotTable Fields Pane. These boxes help to control how the data is displayed and summarized in the PivotTable.
   a. **Filters**

      Purpose: Allows user to add filters to the entire PivotTable.

   b. **Rows**

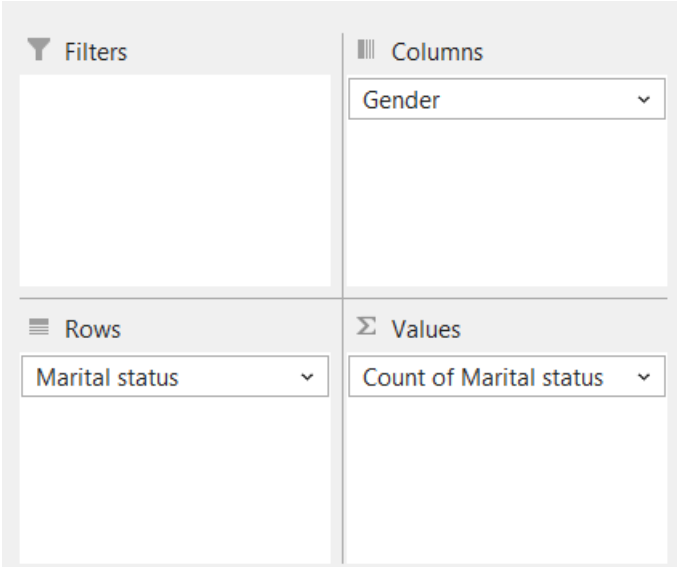
      Purpose: Defines the rows in the PivotTable.

   c. **Columns**

      Purpose: Defines the columns in the PivotTable.

   d. **Values**

      Purpose: Displays the data that is being calculated or summarized.
2. Create a new PivotTable. Follow the steps explained earlier.
3. Drag the field Marital Status to the Rows box, Gender into Columns box and Marital Status (Count) into the Values box (as below)



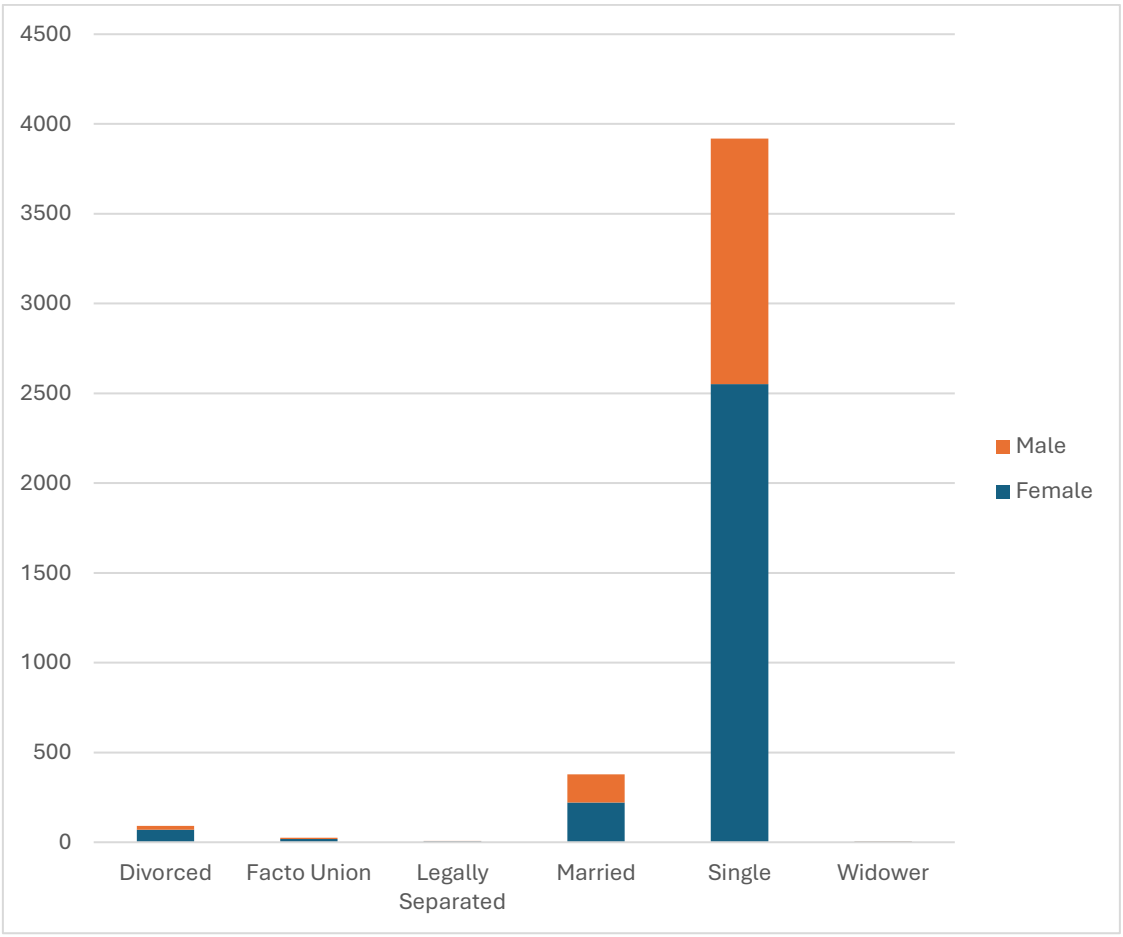
4. A PivotTable (as below) will be displayed.

Count of Marital status	Column Labels		
Row Labels	Female	Male	Grand Total
Divorced	69	22	91
Facto Union	18	7	25
Legally Separated	5	1	6
Married	221	158	379
Single	2552	1367	3919
Widower	3	1	4
Grand Total	**2868**	**1556**	**4424**

5. While selecting the content of the PivotTable, go to the Insert Table and choose Stacked Column in the 2D Column Chart under the Charts group.



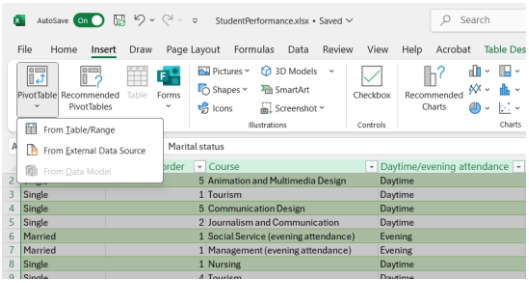
6. The chart (as below) will be created.



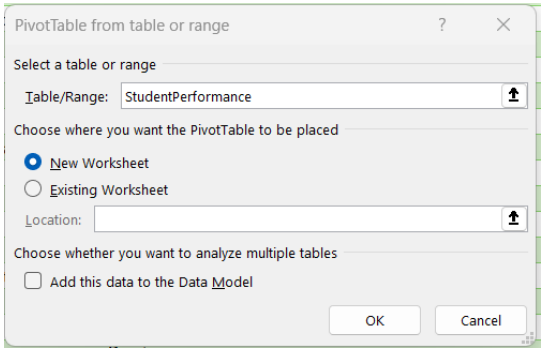
7. Clean up this chart so that it will not have unnecessary information.

Creating Histogram using PivotTable

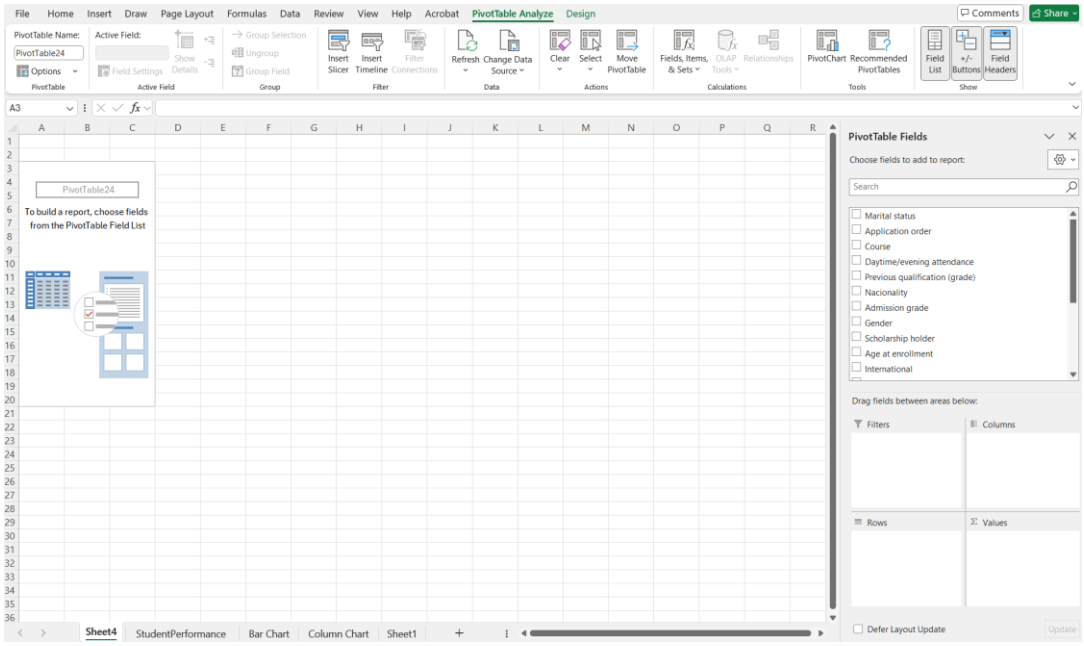
1. Press Ctrl+A to select all the cells that contains data in the StudentPerformance tab.
2. Then, click at the Insert tab and select Pivot Table under the Tables command.



3. A dialog box will be opened (as below). Click OK.

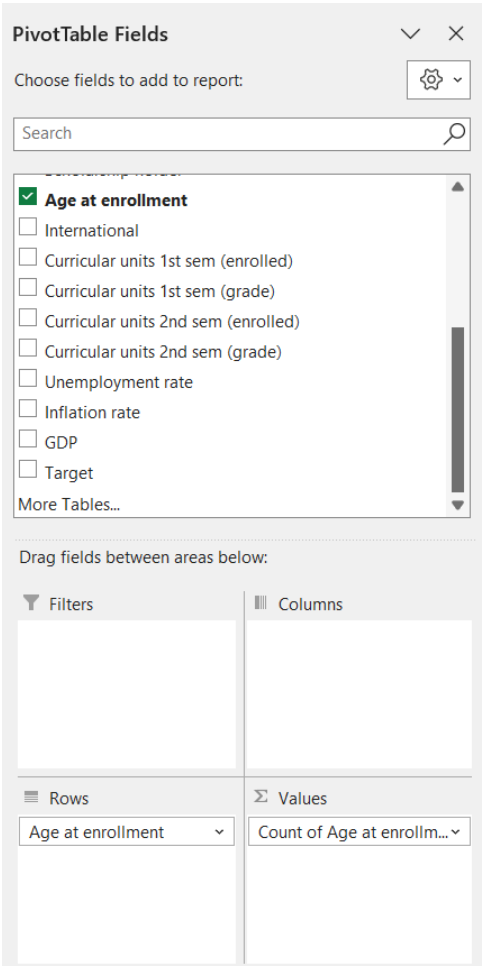


4. A new worksheet will be opened.



5. The PivotTable Fields Pane (on the right) contains all the columns that we have in our dataset.

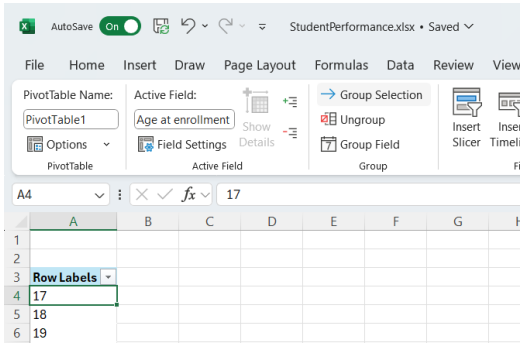
6. Drag Age at enrollment into the Rows box.



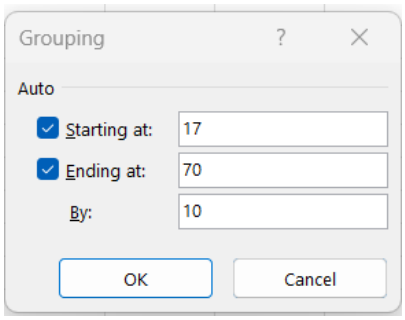
7. A PivotTable will be created.

Row Labels	Count of Age at enrollment
17	5
18	1036
19	911
20	599
21	322
22	174
23	108
24	131
25	93
26	94
27	91
28	83
29	66
30	49
31	55
32	61
33	45
34	60
35	49

8. Go to the Group Selection command in Group.



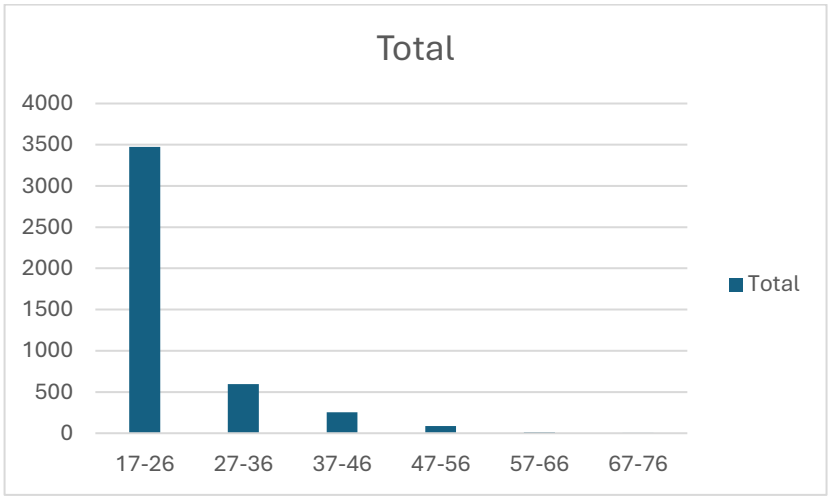
9. A Grouping dialog box will be opened (as below). Fill in the values as shown in the dialog box below. Then, click the OK button.



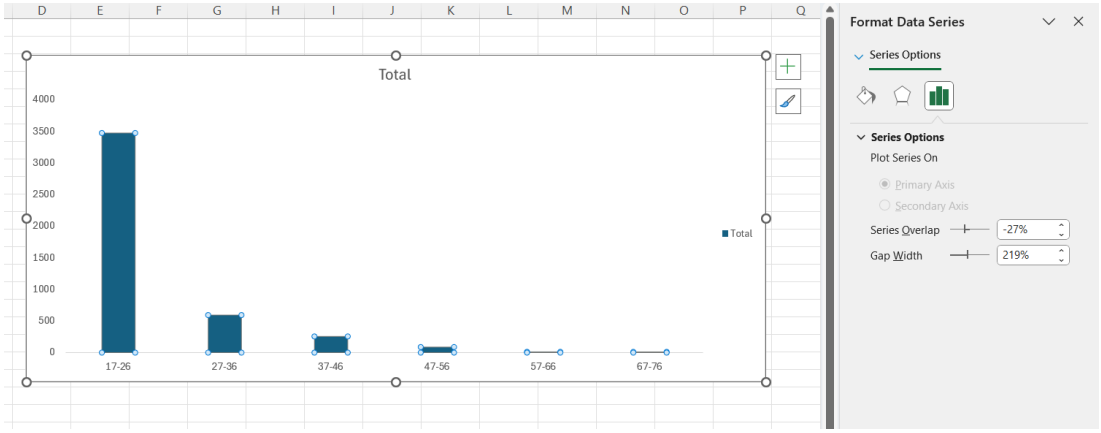
10. The PivotTable (as below) will be created based on the grouping defined earlier.

Row Labels	Count of Age at enrollment
17-26	3473
27-36	594
37-46	256
47-56	88
57-66	12
67-76	1
Grand Total	**4424**

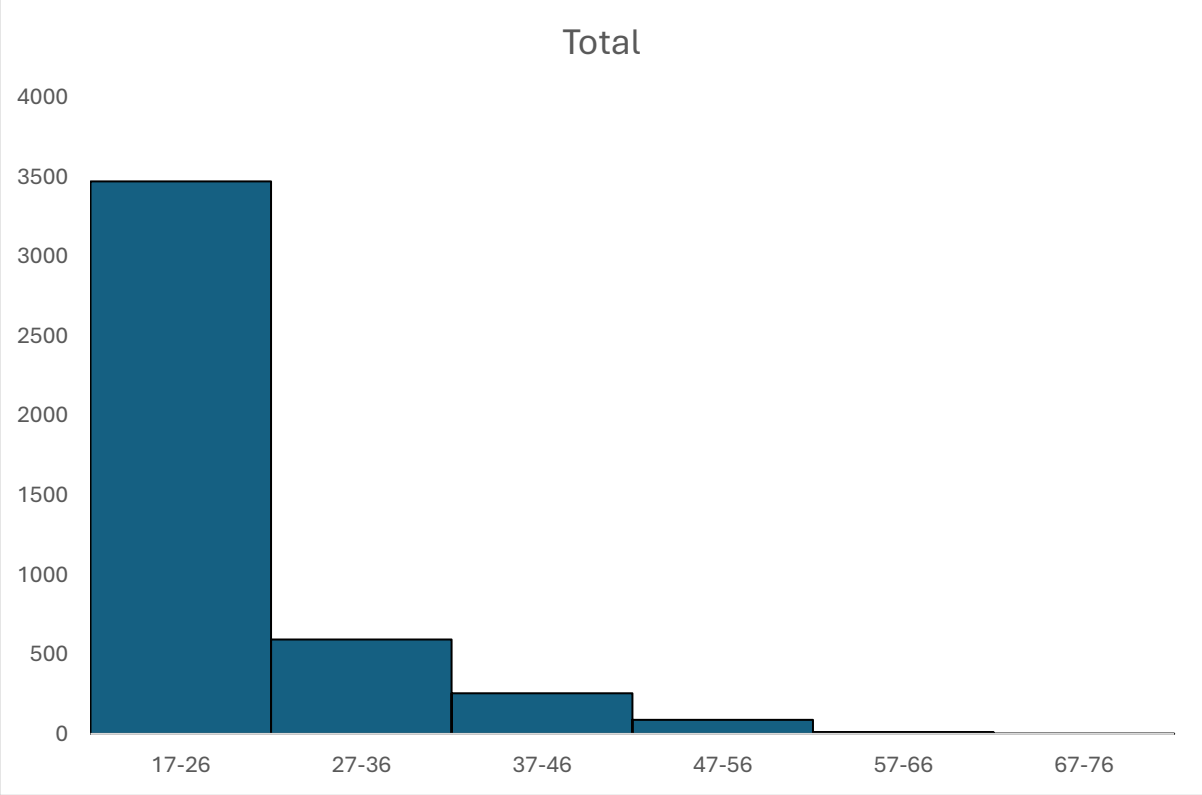
11. Then follow the steps to create a bar chart till you get a bar chart similar to the one shown below. Remove the unnecessary ink in the chart.



12. Mouse over to the bar in the column chart to open up the Format Data Series Pane.

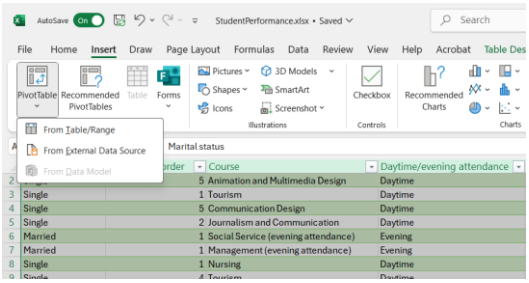


13. Edit the value in the Gap Width to become 0. This will remove the gaps between the bars in the column chart. A new bar chart (as below) which is reflecting the histogram is created.

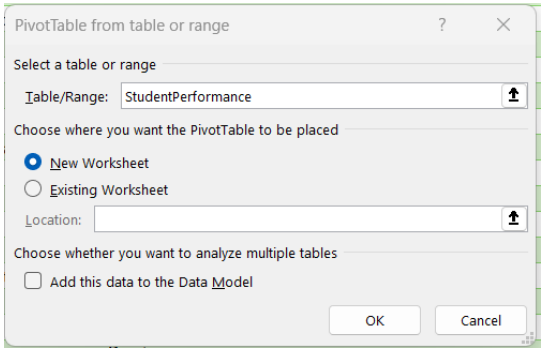


Creating Heatmap using PivotTable

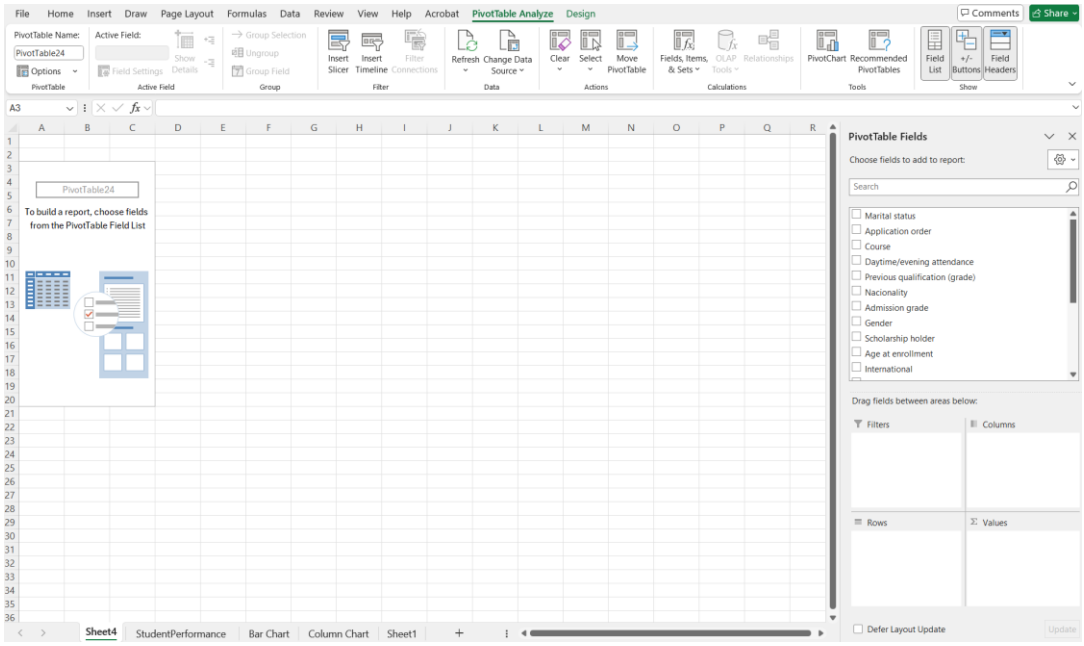
1. Press Ctrl+A to select all the cells that contains data in the StudentPerformance tab.
2. Then, click at the Insert tab and select Pivot Table under the Tables command.



3. A dialog box will be opened (as below). Click OK.

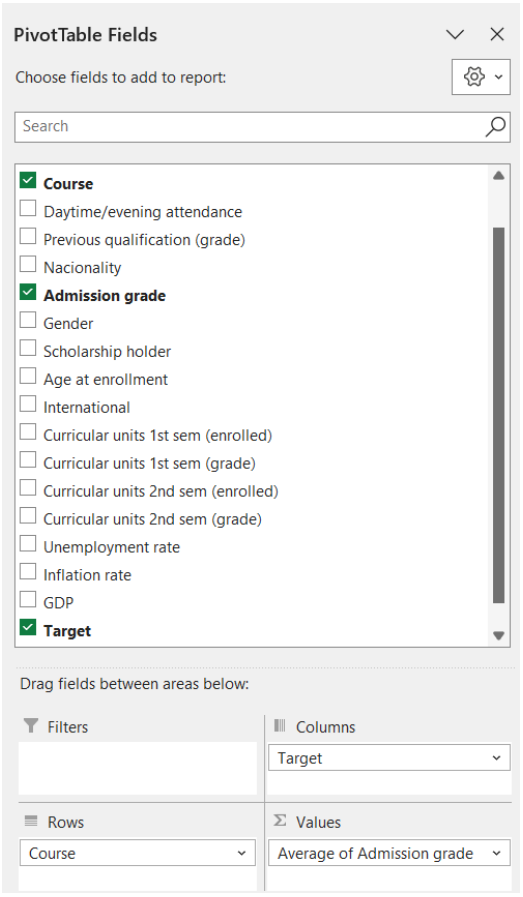


4. A new worksheet will be opened.



5. The PivotTable Fields Pane (on the right) contains all the columns that we have in our dataset.

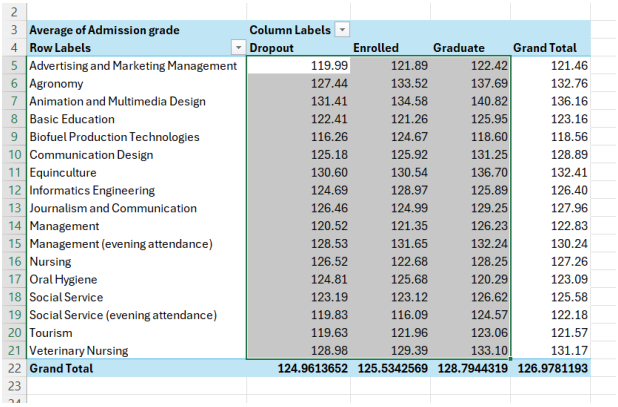
6. Drag Course into the Rows box, Target into the Columns box and Admission grade into the Values box. Ensure that the Values box calculate the average Admission grade.



7. A PivotTable will be created.

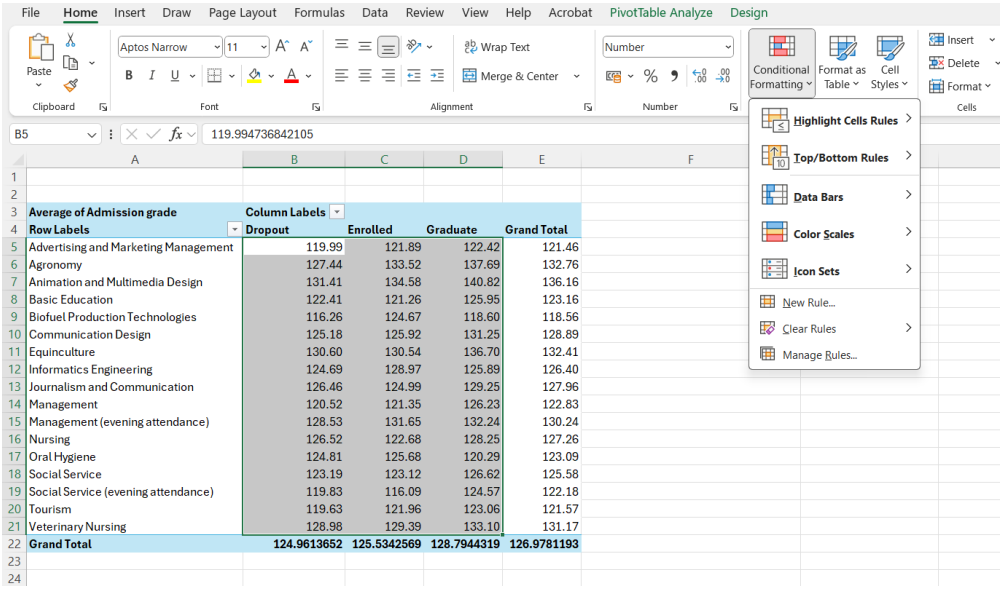
Average of Admission grade	Column Labels			
Row Labels	Dropout	Enrolled	Graduate	Grand Total
Advertising and Marketing Management	119.99	121.89	122.42	121.46
Agronomy	127.44	133.52	137.69	132.76
Animation and Multimedia Design	131.41	134.58	140.82	136.16
Basic Education	122.41	121.26	125.95	123.16
Biofuel Production Technologies	116.26	124.67	118.60	118.56
Communication Design	125.18	125.92	131.25	128.89
Equinculture	130.60	130.54	136.70	132.41
Informatics Engineering	124.69	128.97	125.89	126.40
Journalism and Communication	126.46	124.99	129.25	127.96
Management	120.52	121.35	126.23	122.83
Management (evening attendance)	128.53	131.65	132.24	130.24
Nursing	126.52	122.68	128.25	127.26
Oral Hygiene	124.81	125.68	120.29	123.09
Social Service	123.19	123.12	126.62	125.58
Social Service (evening attendance)	119.83	116.09	124.57	122.18
Tourism	119.63	121.96	123.06	121.57
Veterinary Nursing	128.98	129.39	133.10	131.17
Grand Total	**124.9613652**	**125.5342569**	**128.7944319**	**126.9781193**

8. Select the cells containing the values for Advertising and Marketing Management under the Dropout column until the Veterinary Nursing under the Graduate column.



Average of Admission grade	Column Labels			
Row Labels	Dropout	Enrolled	Graduate	Grand Total
Advertising and Marketing Management	119.99	121.89	122.42	121.46
Agronomy	127.44	133.52	137.69	132.76
Animation and Multimedia Design	131.41	134.58	140.82	136.16
Basic Education	122.41	121.26	125.95	123.16
Biofuel Production Technologies	116.26	124.67	118.60	118.56
Communication Design	125.18	125.92	131.25	128.89
Equinculture	130.60	130.54	136.70	132.41
Informatics Engineering	124.69	128.97	125.89	126.40
Journalism and Communication	126.46	124.99	129.25	127.96
Management	120.52	121.35	126.23	122.83
Management (evening attendance)	128.53	131.65	132.24	130.24
Nursing	126.52	122.68	128.25	127.26
Oral Hygiene	124.81	125.68	120.29	123.09
Social Service	123.19	123.12	126.62	125.58
Social Service (evening attendance)	119.83	116.09	124.57	122.18
Tourism	119.63	121.96	123.06	121.57
Veterinary Nursing	128.98	129.39	133.10	131.17
Grand Total	**124.9613652**	**125.5342569**	**128.7944319**	**126.9781193**

9. Then, on the Home tab, go to Conditional Formatting command under the Styles group. In the Drop-Down menu, choose Color Scales and choose the Green-Yellow-Red scales.



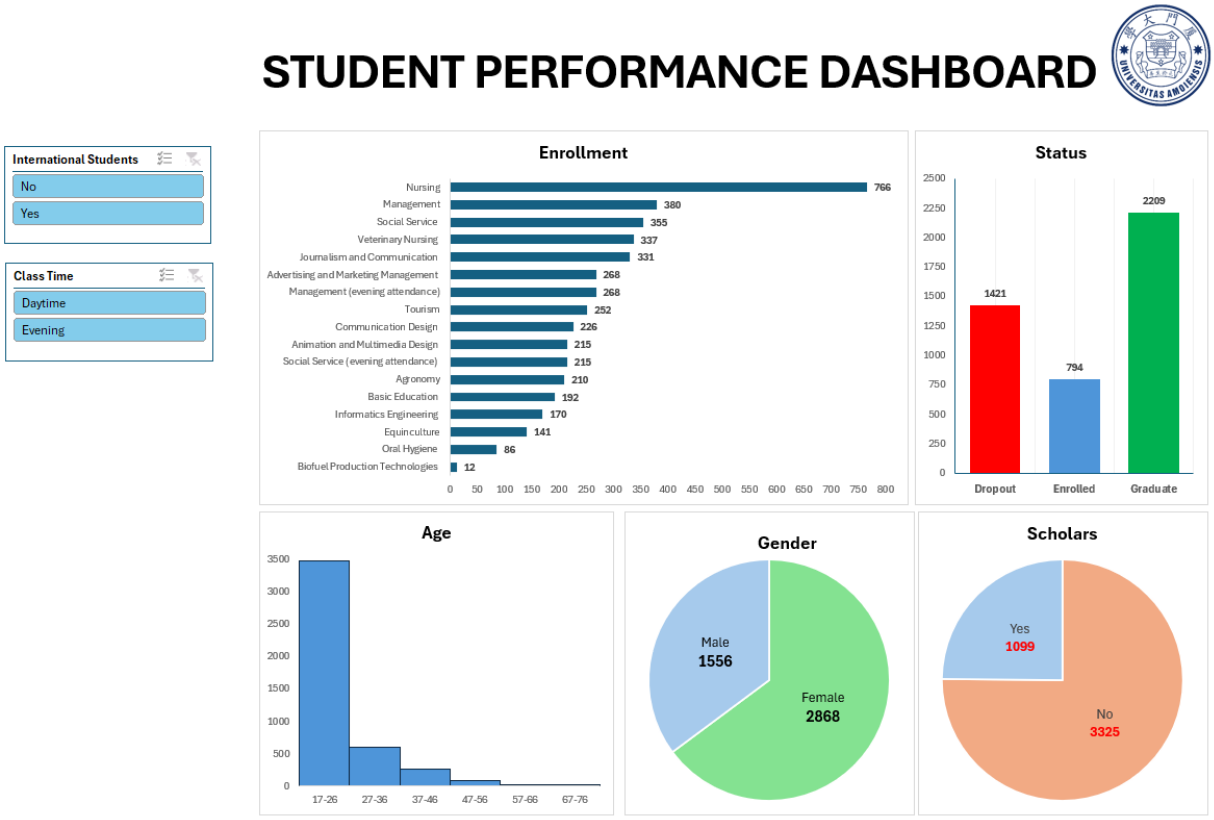
Average of Admission grade	Column Labels			
Row Labels	Dropout	Enrolled	Graduate	Grand Total
Advertising and Marketing Management	119.99	121.89	122.42	121.46
Agronomy	127.44	133.52	137.69	132.76
Animation and Multimedia Design	131.41	134.58	140.82	136.16
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Biofuel Production Technologies	116.26	124.67	118.60	118.56
Communication Design	125.18	125.92	131.25	128.89
Equinculture	130.60	130.54	136.70	132.41
Informatics Engineering	124.69	128.97	125.89	126.40
Journalism and Communication	126.46	124.99	129.25	127.96
Management	120.52	121.35	126.23	122.83
Management (evening attendance)	128.53	131.65	132.24	130.24
Nursing	126.52	122.68	128.25	127.26
Oral Hygiene	124.81	125.68	120.29	123.09
Social Service	123.19	123.12	126.62	125.58
Social Service (evening attendance)	119.83	116.09	124.57	122.18
Tourism	119.63	121.96	123.06	121.57
Veterinary Nursing	128.98	129.39	133.10	131.17
Grand Total	**124.9613652**	**125.5342569**	**128.7944319**	**126.9781193**

10. A heatmap as below will be created.

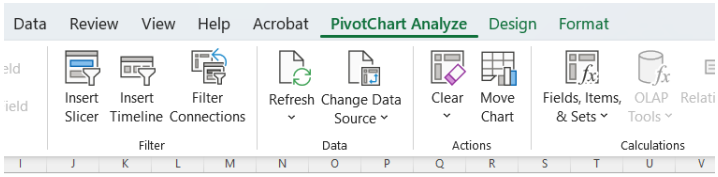
Average of Admission grade	Column Labels			
Row Labels	Dropout	Enrolled	Graduate	Grand Total
Advertising and Marketing Management	119.99	121.89	122.42	121.46
Agronomy	127.44	133.52	137.69	132.76
Animation and Multimedia Design	131.41	134.58	140.82	136.16
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Social Service	123.19	123.12	126.62	125.58
Social Service (evening attendance)	119.83	116.09	124.57	122.18
Tourism	119.63	121.96	123.06	121.57
Veterinary Nursing	128.98	129.39	133.10	131.17
Grand Total	**124.9613652**	**125.5342569**	**128.7944319**	**126.9781193**

Creating Data Dashboard in Excel

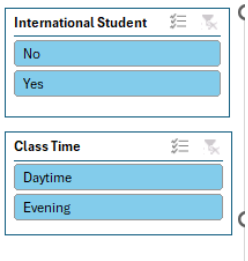
We would like to create a data dashboard as below using all the different charts that we have created earlier.



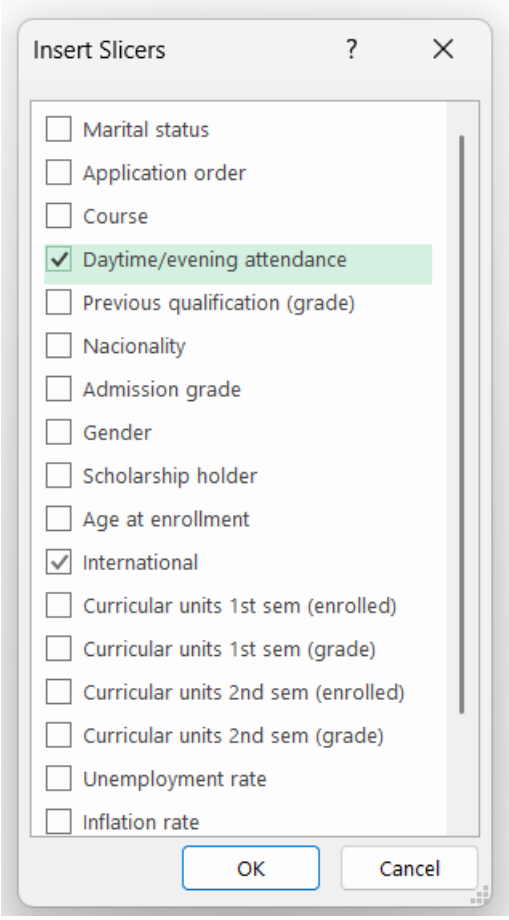
1. Open up a new worksheet in Excel.
2. Copy all the charts that you have created earlier into this new worksheet.
3. Go to the PivotChart Analyze tab and click at the Insert Slicer in the Filter command.



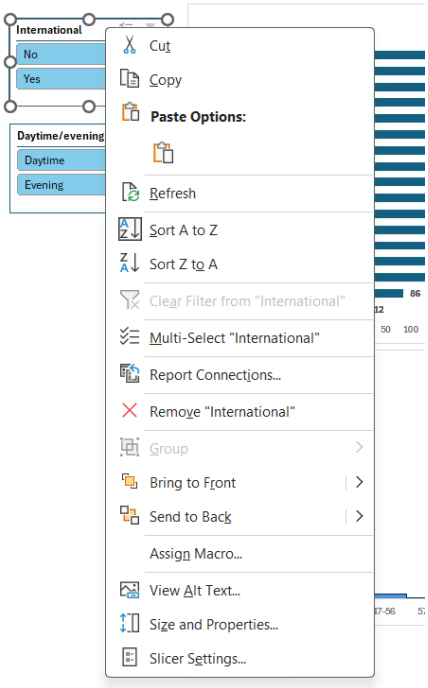
4. We are going to create two (2) slicers in this dashboard.



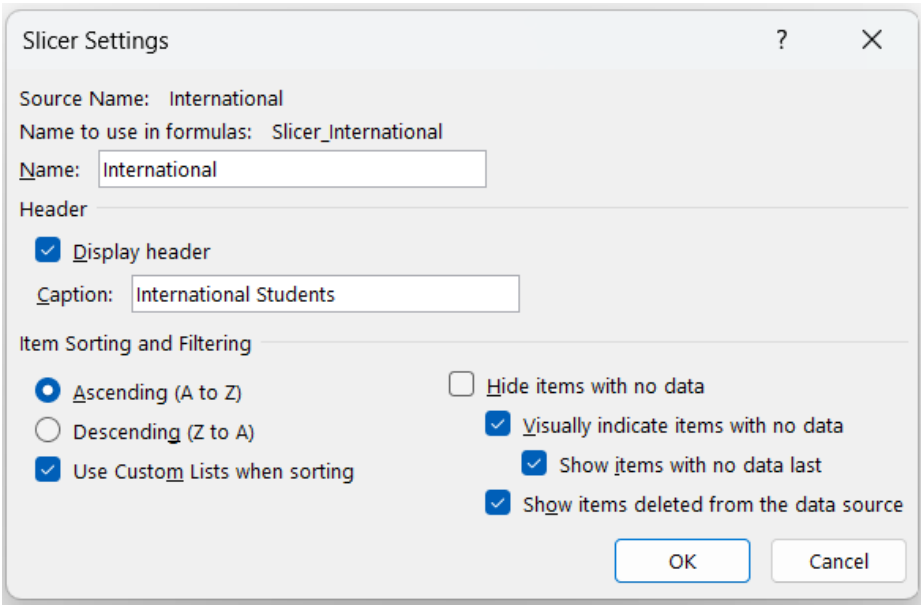
5. An Insert Slicers dialog box will be opened as below. Choose Daytime/evening attendance and International.



6. Right-click at the International slicer and choose Slicer Settings.



7. In the Slicer Setting dialog box, change the caption for the Display to International Students.



8. Change the caption for Daytime/evening attendance to Class Time. Follow the steps in Step 6 and 7.